Fusion draft: SST-28/46/50/66 Time and relational canon

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About this fusion draft

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Chapter 1

SST-28 — Time from Swirl

Time from Swirl: A Hydrodynamic Origin of Proper Time

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Abstract

Swirl–String Theory (SST) models matter as knotted swirl strings embedded in an incompressible, inviscid swirl condensate of effective density ρ_f. In this paper we develop a swirl–clock Hamiltonian and Lagrangian formulation in which internal time evolution is controlled by the local tangential swirl speed rather than by spacetime curvature. At the coarse–grained level, the condensate dynamics are described by a velocity–vorticity pair $(\vec{v}, \boldsymbol{\omega} = \nabla \times \vec{v})$ and a local swirl energy density

$$\mathcal{H}_{\text{swirl}} = \frac{1}{2}\rho_m\Big(|\vec{v}|^2 + \ell_\omega^2|\boldsymbol{\omega}|^2\Big), \qquad \ell_\omega \sim r_c,$$

which encodes bulk kinetic energy and a short–range swirl tension. The associated swirl clock is introduced as

$$S_{(t)}^{\circlearrowleft}(\vec{x}) = \sqrt{1 - \frac{|v_\perp(\vec{x})|^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}}, \qquad dt(\vec{x}) = dt_\infty\, S_{(t)}^{\circlearrowleft}(\vec{x}),$$

leading to a "redshifted" Schrödinger evolution $i\hbar\,\partial_{t_\infty}\Psi = S_{(t)}^{\circlearrowleft}(\vec{x})\,H[\vec{v}]\,\Psi$ for swirl–bound modes. Starting from a local, time–symmetric Lagrangian density $\mathcal{L}_{\text{swirl}} = \frac{1}{2}\rho_m|\vec{v}|^2 - \frac{1}{2}\rho_m\ell_\omega^2|\nabla\times\vec{v}|^2$, we derive a vector Helmholtz equation $\nabla^2\vec{v} + \ell_\omega^{-2}\vec{v} = 0$ whose bound solutions define discrete swirl shells and an effective core radius $\sim \ell_\omega$. A symmetric swirl stress tensor is constructed, providing an energy–momentum bookkeeping scheme in which swirl tension plays the role of a hydrodynamic analogue of magnetic stress. Finally, we introduce an effective metric ansatz

$$ds^2 = -\Big(1 - \tfrac{v_\perp^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}\Big)\,dt^2 + \big(\delta_{ij} + \gamma_{ij}(\boldsymbol{\omega})\big)\,dx^i dx^j,$$

with $\gamma_{ij} \propto \omega_i\omega_j - \frac{1}{2}\delta_{ij}|\boldsymbol{\omega}|^2$, to summarize the impact of swirl on internal clocks and rulers while preserving a flat background condensate. Using the canonical SST constants and an axisymmetric reference profile, we obtain finite, strongly peaked swirl energy densities and a core swirl clock $S_{(t)}^{\circlearrowleft}(r_c) \approx 0.93$, providing a quantitatively controlled link between swirl structure, time dilation, and emergent geometry within the Rosetta 0.6.0 framework.

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1.1 Swirl–Clock Hamiltonian and Time Dilation in Swirl–String Theory

In Swirl–String Theory (SST), matter is modeled as knotted swirl strings embedded in a mechanically incompressible, inviscid swirl condensate of effective density ρ_f. The internal time evolution of a bound excitation is governed not by spacetime curvature, but by the rotational kinetic energy stored in structured swirl. Building on classical vortex kinematics [1, 3, 4, 11, 6] and using a Schrödinger-type evolution for internal modes [2], we construct a swirl–clock Hamiltonian in which the local clock rate is set by the tangential swirl speed.

1.1.1 Velocity–Vorticity Kinematics and Swirl Energy Density

Let $\vec{v}(\vec{x},t)$ be the local condensate velocity and

$$\boldsymbol{\omega}(\vec{x},t) := \nabla \times \vec{v}(\vec{x},t) \tag{1.1}$$

the associated vorticity. At the coarse–grained level used in Rosetta 0.6.0, the leading local energy density retained in the effective Hamiltonian is

$$\mathcal{H}_{\text{swirl}}(\vec{x}) = \frac{1}{2}\,\rho_m\,\Big(|\vec{v}(\vec{x})|^2 + \ell_\omega^2\,|\boldsymbol{\omega}(\vec{x})|^2\Big), \qquad \ell_\omega \sim r_c, \tag{1.2}$$

where ρ_m is the mass–equivalent density associated with the core of the swirl string. The $|\vec{v}|^2$ term is the bulk kinetic energy density; the $\ell_\omega^2|\boldsymbol{\omega}|^2$ term is the leading local surrogate for the nonlocal Biot–Savart interaction energy and encodes a short–distance swirl tension [11, 6, 7, 8].

Dimensions:

$$[\rho_m][v]^2 \sim \mathrm{J/m^3}, \qquad [\rho_m]\ell_\omega^2[\omega]^2 \sim \mathrm{J/m^3},$$

so $\mathcal{H}_{\text{swirl}}$ is an energy density.

Axisymmetric profile (Rosetta reference ansatz). For explicit estimates we use the axisymmetric angular–velocity profile

$$\Omega_{\text{swirl}}(r) = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}\,e^{-r/r_c}, \qquad v_\theta(r) = r\,\Omega_{\text{swirl}}(r) = \|\mathbf{v}_{\circlearrowleft}\|\Big(\frac{r}{r_c}\Big)e^{-r/r_c}, \tag{1.3}$$

where $\|\mathbf{v}_{\circlearrowleft}\|$ is the characteristic swirl speed ($\|\mathbf{v}_{\circlearrowleft}\| \approx 1.09384563 \times 10^6$ m/s in the SST Canon). The tangential speed peaks at $r = r_c$ with

$$v_\theta(r_c) = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{e}. \tag{1.4}$$

The corresponding axial vorticity is

$$\omega_z(r) = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\big(r\,v_\theta(r)\big) = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}e^{-r/r_c}\Big(2 - \frac{r}{r_c}\Big). \tag{1.5}$$

1.1.2 Swirl–Clock and Proper Time from Tangential Swirl

Rosetta 0.6.0 defines the local swirl clock $S_{(t)}^{\circlearrowleft}(\vec{x})$ as a dimensionless factor that slows internal evolution relative to an asymptotic laboratory time t_∞. At leading order, the swirl clock is taken to depend on the local tangential swirl speed $v_\perp(\vec{x})$ through

$$dt(\vec{x}) = dt_\infty\,S_{(t)}^{\circlearrowleft}(\vec{x}), \qquad S_{(t)}^{\circlearrowleft}(\vec{x}) := \sqrt{1 - \frac{|v_\perp(\vec{x})|^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}} \in [0,1], \tag{1.6}$$

in direct analogy with special–relativistic time dilation, but with $\|\mathbf{v}_{\circlearrowleft}\|$ replacing c as the relevant swirl scale. For an axisymmetric swirl string, $v_\perp = v_\theta\,\hat{\boldsymbol{\theta}}$. Inserting (1.3) gives

$$S^{\circlearrowleft}_{(t)}(r) = \sqrt{1 - \left(\frac{r}{r_c}\right)^2 e^{-2r/r_c}}, \qquad S^{\circlearrowleft}_{(t)}(r_c) = \sqrt{1 - e^{-2}} \approx 0.929873. \tag{1.7}$$

Thus the core clock runs at roughly 93% of the far–field rate for the canonical profile.

1.1.3 Hamiltonian Time Evolution of a Swirl–Bound Mode

Let Ψ denote a swirl–bound internal mode supported on a given knot–type swirl string. In local proper time t_{loc} one has the usual Schrödinger form

$$i\hbar\,\frac{\partial \Psi}{\partial t_{\mathrm{loc}}} = H[\vec{v}]\,\Psi, \tag{1.8}$$

where $H[\vec{v}]$ is constructed from the energy functional associated with $\mathcal{H}_{\mathrm{swirl}}$ in (1.2). Since $dt_{\mathrm{loc}} = S^{\circlearrowleft}_{(t)}(\vec{x})\,dt_\infty$, the evolution in the laboratory time t_∞ is

$$i\hbar\,\frac{\partial}{\partial t_\infty}\Psi(\vec{x}, t_\infty) = S^{\circlearrowleft}_{(t)}(\vec{x})\,H[\vec{v}]\,\Psi(\vec{x}, t_\infty). \tag{1.9}$$

For a single–mode reduction localized in the knot core, we approximate

$$\langle S^{\circlearrowleft}_{(t)}\rangle := \frac{1}{V}\int_V S^{\circlearrowleft}_{(t)}(\vec{x})\,\mathrm{d}^3x, \qquad i\hbar\,\frac{\mathrm{d}}{\mathrm{d}t_\infty}\Psi = \langle S^{\circlearrowleft}_{(t)}\rangle\,H[\vec{v}]\,\Psi. \tag{1.10}$$

Dimensional check. In (1.6), the argument of the square root is dimensionless. In (1.2), each term carries units $\mathrm{J/m^3}$. Equation (1.9) matches the usual "redshifted Hamiltonian" form where the local clock factor multiplies H.

1.1.4 Numerical validation with SST Canon constants

Using the SST Canon values $\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6\,\mathrm{m/s}$, $r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}$, and $\rho_m = 3.8934358266918687 \times 10^{18}\,\mathrm{kg/m^3}$, and the profile (1.3)–(1.5), one finds

$$\begin{aligned}
v_\theta(r_c) &= \frac{\|\mathbf{v}_{\circlearrowleft}\|}{e} = 4.02403 \times 10^5\,\mathrm{m/s},\\
S^{\circlearrowleft}_{(t)}(r_c) &= \sqrt{1 - e^{-2}} = 0.9298734950,\\
\omega_z(r_c) &= \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} e^{-1}(2-1) = 2.8560102099 \times 10^{20}\,\mathrm{s^{-1}},\\
\mathcal{H}_{\mathrm{swirl}}(r_c) &= \frac{1}{2}\rho_m\left[v_\theta(r_c)^2 + r_c^2\omega_z(r_c)^2\right] = 6.3045795546 \times 10^{29}\,\mathrm{J/m^3}.
\end{aligned}$$

Representative values for several radii $r/r_c \in \{0, 0.25, 0.5, 1, 2, 3\}$ are summarized in Table 1.1, using the same Canon constants.

1.2 Cauchy Integral Theorem and Swirl–Clock Plateaus

The swirl–clock Hamiltonian in Sect. 1.1 can be sharpened by using the complex–analytic structure of incompressible planar flows. In the transverse plane to a straight segment of a swirl string, the swirl condensate velocity is divergence-free and (outside the core) irrotational, so it admits a complex potential in the sense of classical hydrodynamics [10, 3, 4, 11].

Table 1.1: Representative profile values for the canonical swirl string ($\|\mathbf{v}_{\circlearrowleft}\|, r_c, \rho_m$ from SST Canon).

r/r_c	r (m)	v_θ (m/s)	$S^{\circlearrowleft}_{(t)}$	ω_z (s^{-1})	$\mathcal{H}_{\text{swirl}}$ (J/m^3)
0.00	0	0	1.0000000000	$1.5526881311 \times 10^{21}$	$9.3169784018 \times 10^{30}$
0.25	3.5224×10^{-16}	2.12972×10^{5}	0.9808628007	$1.0580803908 \times 10^{21}$	$4.4148695754 \times 10^{30}$
0.50	7.0449×10^{-16}	3.31725×10^{5}	0.9529061547	$7.0631471735 \times 10^{20}$	$2.1422030049 \times 10^{30}$
1.00	1.4090×10^{-15}	4.02403×10^{5}	0.9298734950	$2.8560102099 \times 10^{20}$	$6.3045795546 \times 10^{29}$
2.00	2.8179×10^{-15}	2.96072×10^{5}	0.9626720337	0	$1.7064641194 \times 10^{29}$
3.00	4.2269×10^{-15}	1.63378×10^{5}	0.9887827013	$-3.8651895068 \times 10^{19}$	$5.7736201234 \times 10^{28}$

1.2.1 Complex swirl potential and circulation

Consider a straight swirl string aligned with the z–axis, and introduce the complex coordinate

$$z = x + iy, \tag{1.11}$$

in a transverse cross–section. In a simply connected region that does not contain the core line, there exists a complex potential

$$W(z) = \Phi(x, y) + i\Psi(x, y), \tag{1.12}$$

such that the complex velocity

$$U(z) := v_x - iv_y = \frac{dW}{dz} \tag{1.13}$$

is analytic [10]. The (scalar) circulation around a closed loop C in the x–y plane is

$$\Gamma_C := \oint_C \vec{v} \cdot d\vec{\ell} = \oint_C (v_x\, dx + v_y\, dy) = -\,\text{Im} \oint_C U(z)\, dz. \tag{1.14}$$

If $U(z)$ is analytic in the region bounded by C, Cauchy's integral theorem implies

$$\oint_C U(z)\, dz = 0 \quad \Rightarrow \quad \Gamma_C = 0, \tag{1.15}$$

so any loop that does not link the core measures negligible circulation.

When the swirl string axis is treated as an isolated singular line (removed from the fluid domain), the complex velocity $U(z)$ is analytic on the punctured plane and admits a Laurent expansion with a simple pole at $z = 0$. By Cauchy's integral and residue theorems [11, 10],

$$\oint_C U(z)\, dz = 2\pi i\, n\, \text{Res}(U, z = 0)\,, \tag{1.16}$$

where $n \in \mathbb{Z}$ is the winding number of C about the axis. Writing

$$\text{Res}(U, z = 0) = \frac{\kappa}{2\pi i}, \tag{1.17}$$

we obtain the circulation plateau

$$\Gamma_C = -\,\text{Im} \oint_C U(z)\, dz = n\,\kappa, \qquad n \in \mathbb{Z}, \tag{1.18}$$

in agreement with Kelvin's circulation theorem for material loops [1, 11, 6]. Loops whose spanning disk intersects the core annulus form a family with constant Γ_C; loops that do not link the axis yield $\Gamma_C \approx 0$.

1.2.2 Canonical identification of the circulation quantum κ

Within the SST Canon and Rosetta 0.6.0, the circulation quantum is tied to core kinematics by

$$\boxed{\kappa \equiv \frac{h}{m_{\text{eff}}} = 2\pi\, r_c\, \|\mathbf{v}_{\circlearrowleft}\|} \tag{1.19}$$

where r_c is the swirl–string core radius and $\|\mathbf{v}_{\circlearrowleft}\| = \|\mathbf{v}_{\circlearrowleft}\|$ is the characteristic swirl speed. This gives an effective mass

$$m_{\text{eff}} = \frac{h}{\kappa} = \frac{h}{2\pi\, r_c\, \|\mathbf{v}_{\circlearrowleft}\|}. \tag{1.20}$$

Using the Canon values

$$r_c = 1.40897017 \times 10^{-15}\,\text{m}, \qquad \|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^{6}\,\text{m/s},$$

we obtain

$$\kappa = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9}\,\text{m}^2/\text{s}, \qquad m_{\text{eff}} = \frac{h}{\kappa} \approx 6.84 \times 10^{-26}\,\text{kg}, \tag{1.21}$$

which corresponds to an energy scale of $\mathcal{O}(10^1\,\text{GeV})$ when expressed in c^{-2} units. Dimensions are consistent: $[\kappa] = \text{m}^2\text{s}^{-1}$.

1.2.3 Quantized swirl–clock loops

For a circular loop of radius r about the axis, the azimuthal swirl speed is

$$v_\theta^{(n)}(r) = \frac{\Gamma_C}{2\pi r} = \frac{n\,\kappa}{2\pi r}, \qquad n \in \mathbb{Z}, \tag{1.22}$$

so that the swirl clock (1.6) becomes

$$S^{\circlearrowleft}_{(t)n}(r) := \sqrt{1 - \frac{\left(v_\theta^{(n)}(r)\right)^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}} = \sqrt{1 - \frac{n^2\,\kappa^2}{4\pi^2 r^2 \|\mathbf{v}_{\circlearrowleft}\|^2}}. \tag{1.23}$$

In other words, for any fixed radius r in the envelope of the swirl string, the swirl clock takes values on a discrete set indexed by the linking number n. Topology changes that alter n (e.g. reconnections, nucleations, or annihilations of swirl lines) therefore produce $\mathcal{O}(1)$ jumps in the local clock factor, even if the underlying geometry varies smoothly.

In the thin–core limit $r \approx r_c$, Eq. (1.23) ties the minimal clock rate directly to the Canon parameters:

$$S^{\circlearrowleft}_{(t)n=1}(r_c) = \sqrt{1 - \frac{\kappa^2}{4\pi^2 r_c^2 \|\mathbf{v}_{\circlearrowleft}\|^2}} = \sqrt{1 - \frac{1}{4\pi^2}} \approx 0.9603, \tag{1.24}$$

while higher n are suppressed by the requirement that the local swirl speed remains in the physically allowed regime. In practice, the axisymmetric profile (1.3) provides a smoothed version of the thin–core idealization, with the same integer plateau structure inherited from (1.18).

1.2.4 Impact on the swirl–clock Hamiltonian

The quantized circulation plateaus (1.18) and discrete swirl clocks (1.23) refine the Hamiltonian evolution

$$i\hbar\,\partial_{t_\infty}\Psi = S^{\circlearrowleft}_{(t)}(\vec{x})\, H[\vec{v}]\,\Psi \tag{1.25}$$

by promoting $S^{\mho}_{(t)}(\vec{x})$ to a topological observable: for a given family of loops that link a swirl string segment, the effective clock factor entering the single–mode reduction is locked to the integer n. In particular, for a mode localized near $r \approx r_c$,

$$\langle S^{\mho}_{(t)} \rangle \;\longrightarrow\; \langle S^{\mho}_{(t)n=1} \rangle \quad \text{under smooth deformations that preserve the linking number,} \tag{1.26}$$

while topology–changing events produce discrete shifts in $\langle S^{\mho}_{(t)} \rangle$. This is the swirl–clock analogue of the integer plateau of circulation used in the long–distance swirl gravity construction, now embedded directly in the time–dilation sector of SST.

Analogy. In complex analysis language, the swirl string core plays the role of a simple pole: as long as a loop winds around it, Cauchy's theorem fixes the integral, and the clock slows by a fixed amount; only when the loop slips off the pole (a topology change) can the integral, and the clock rate, jump.

1.3 Swirl Lagrangian and Stationary Field Equation

To complement the Hamiltonian formulation, we adopt a local, time–symmetric Lagrangian density for the swirl velocity field,

$$\mathcal{L}_{\text{swirl}} = \frac{1}{2}\rho_m\,|\vec{v}|^2 - \frac{1}{2}\rho_m\,\ell_\omega^2\,|\nabla \times \vec{v}|^2, \qquad \ell_\omega \sim r_c, \tag{1.27}$$

interpreted as the lowest–order local truncation of the nonlocal swirl interaction energy [11, 6]. Varying with respect to $\vec{v}$ yields

$$\rho_m\,\vec{v} - \rho_m\,\ell_\omega^2\,\nabla \times (\nabla \times \vec{v}) = 0. \tag{1.28}$$

For incompressible flow ($\nabla\cdot\vec{v} = 0 \Rightarrow \nabla \times (\nabla \times \vec{v}) = -\nabla^2\vec{v}$) this reduces to the vector Helmholtz equation

$$\boxed{\nabla^2\vec{v} + \frac{1}{\ell_\omega^2}\,\vec{v} = 0} \tag{1.29}$$

whose bound solutions define discrete swirl shells and an effective core radius $\sim \ell_\omega$. Combined with the swirl–clock relation (1.6), any stationary swirl solution $\vec{v}(\vec{x})$ has a well–defined time–dilation profile.

Dimensional check. $\mathcal{L}_{\text{swirl}}$ carries units $\mathrm{J/m^3}$. In (1.29), $1/\ell_\omega^2$ has dimensions of wavenumber squared.

1.4 Swirl Stress Tensor and Energy–Momentum Bookkeeping

For the Lagrangian (1.27), a symmetric Cauchy stress consistent with spatial translation invariance is [9, 8]

$$\sigma_{ij} = \rho_m\,v_i v_j + \rho_m\,\ell_\omega^2\Big(\omega_i\omega_j - \frac{1}{2}\delta_{ij}|\boldsymbol{\omega}|^2\Big) - \delta_{ij}\,\frac{1}{2}\rho_m|\vec{v}|^2, \tag{1.30}$$

where ω_i are the components of $\boldsymbol{\omega}$. The first term is inertial flux, the second encodes swirl tension (closely analogous to magnetic tension in Maxwell theory [3, 11]), and the last term is the kinetic pressure contribution (bulk pressure may be absorbed into $-\delta_{ij}p$).

The corresponding energy density and energy flux are

$$\mathcal{H}_{\text{swirl}} = \frac{1}{2}\rho_m\,|\vec{v}|^2 + \frac{1}{2}\rho_m\,\ell_\omega^2|\boldsymbol{\omega}|^2, \qquad \boldsymbol{\mathcal{S}}_{\text{energy}} \sim (\mathcal{H}_{\text{swirl}} + p)\,\vec{v}, \tag{1.31}$$

in direct analogy with ideal–fluid and magnetohydrodynamic energy fluxes [9, 4, 7].

For bookkeeping in an effective $3 + 1$ description, one may define a swirl energy–momentum tensor

$$T^{00}_{(\mathrm{swirl})} := \mathcal{H}_{\mathrm{swirl}}, \quad T^{0i}_{(\mathrm{swirl})} := \rho_m v^i, \quad T^{ij}_{(\mathrm{swirl})} := \sigma_{ij}, \tag{1.32}$$

so that time dilation, inertial response, and effective "gravitational" behaviour are all traced back to the distribution of $\mathcal{H}_{\mathrm{swirl}}$ and σ_{ij} within and around a swirl string.

1.5 Emergent Metric Bookkeeping from Swirl Foliations

SST maintains an underlying Euclidean spatial background and a global laboratory time t_∞. Nevertheless, it is often convenient to represent the combined effect of the swirl clock and swirl tension as an *effective* metric that tracks how internal processes perceive time and spatial distances.

1.5.1 Temporal Direction from the Swirl–Null Line

Consider a knotted swirl string (e.g. an electron–like or proton–like excitation) supported by a localized region of high vorticity $\boldsymbol{\omega}$. Define the *swirl–null line* $\mathcal{N} \subset \mathbb{R}^3$ as the one–dimensional locus

$$\boldsymbol{\omega}(\vec{x}) = 0, \qquad \nabla\cdot\boldsymbol{\omega}(\vec{x}) = 0, \qquad \vec{x} \in \mathcal{N}. \tag{1.33}$$

Along $\mathcal{N}$ the swirl string neither stores swirl energy nor exhibits swirl–induced time dilation. The unit 4–vector tangent to $\mathcal{N}$,

$$\tau^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}s}\Big|_{\boldsymbol{\omega}=0}, \tag{1.34}$$

may therefore be used as an internal temporal axis for the excitation: it is the direction of maximal proper–time flow within the knot.

1.5.2 Spatial Layers from Swirl Shells

The region surrounding $\mathcal{N}$ is foliated by nested isovorticity shells

$$\Sigma_\lambda := \{\vec{x} \in \mathbb{R}^3 \mid |\boldsymbol{\omega}(\vec{x})| = \lambda\}, \tag{1.35}$$

each of which carries a swirl clock

$$dt(\lambda) = dt_\infty \sqrt{1 - \frac{v_\perp(\lambda)^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}} \tag{1.36}$$

set by the local geometry of $\vec{v}$ or $\boldsymbol{\omega}$. In this picture, spatial distance from the swirl–null line is encoded in the radial variation of swirl intensity and the associated layering of clock rates.

A dimensionless spatial correction consistent with isotropy is

$$g_{ij}(\vec{x}) = \delta_{ij} + \frac{\ell_g^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}\Big(\omega_i\omega_j - \frac{1}{2}\delta_{ij}|\boldsymbol{\omega}|^2\Big), \qquad \ell_g \sim r_c, \tag{1.37}$$

so that the effective line element reads

$$ds^2 = -\Big(1 - \frac{v_\perp^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}\Big)\, dt^2 + \left(\delta_{ij} + \gamma_{ij}(\boldsymbol{\omega})\right) dx^i dx^j, \quad \gamma_{ij}(\boldsymbol{\omega}) := \frac{\ell_g^2}{\|\mathbf{v}_{\circlearrowleft}\|^2}\Big(\omega_i\omega_j - \frac{1}{2}\delta_{ij}|\boldsymbol{\omega}|^2\Big). \tag{1.38}$$

This metric is an internal bookkeeping device: it summarizes how internal clocks and rulers are distorted by the swirl configuration, while the fundamental kinematics of SST remain those of a flat background condensate.

One–line analogy

A swirl string is like a spinning rubber band in water: the faster its surface water flows, the slower the clock at its centre ticks.

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Chapter 2

SST-46 — Relational time of arrival

[Relational Time-of-Arrival]Time from Swirl: A Hydrodynamic Origin of Proper Time

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Abstract

Time-of-arrival (TOA) in quantum theory sits at an awkward intersection: Pauli-type arguments obstruct a self-adjoint operator canonically conjugate to a semibounded Hamiltonian, while relativistic localization makes "the" time parameter detector- and frame-dependent. We propose a covariant alternative that treats TOA as a relational field observable rather than an operator. The construction uses two conserved ingredients: a matter current J^μ that records detector crossings through a world-tube Σ, and an event-count current $j^\mu_{\rm ev}$ whose integral defines a discrete clock $N_{\rm ev}$. Coarse-graining $j^\mu_{\rm ev}$ over a scale ℓ yields an infrared clock field $T(x)$, and TOA becomes the flux through Σ conditioned on the local clock reading $T|_\Sigma$. In a $1+1$-dimensional example with a narrowband Klein–Gordon wavepacket and a gapped quadratic clock EFT, the TOA density reduces to a convolution of the semiclassical flux profile with the clock readout distribution. This predicts a calculable arrival-time broadening set by the clock variance, while massive clock correlations impose an exponentially suppressed envelope on contributions that would require spacelike clock correlations. The result is a TOA observable whose dependence on detector details is compressed into the geometry of Σ, the effective flux J^μ, and the clock-sector correlators, together with a controlled UV-to-IR matching between event counts and continuum time.

relational time, time of arrival, quantum time observables, covariant observables, event currents, quantum field theory

Notation summary

J^μ	conserved matter current (detector flux)
$j^\mu_{\rm ev}$	conserved event-count current (clock sector)
$N_{\rm ev}$	discrete event-count clock
$T(x)$ or $\chi(x)$	continuum clock field (IR limit)
Σ	detector world-tube
$\mathcal{F}$	detector flux functional $n_\mu J^\mu$
$(\cdot)_\ell$	coarse-graining over spacetime scale ℓ

2.1 Introduction

Time is usually supplied to quantum theory as an external parameter. Operationally, however, time is read from physical degrees of freedom, and in relativistic settings those degrees of freedom are local. This mismatch is visible already in nonrelativistic quantum mechanics: Pauli-type objections constrain the existence of a self-adjoint time operator canonically conjugate to a semibounded Hamiltonian in generic settings [1, 4]. In quantum field theory the situation is sharper, because localization, causality, and detector modelling enter from the start [4, 5].

Many successful TOA constructions therefore adopt explicitly operational definitions: POVM-based prescriptions such as Kijowski's [2] and later developments [3, 4], as well as detector models and measurement couplings [5]. These approaches produce useful arrival-time statistics, but typically tie the observable to a chosen clock variable, a preferred slicing, or detector microphysics.

In this paper we take a different starting point. We define TOA as a *relational field observable*: detector crossings are binned by a *physical clock field* constructed from a conserved event-count current. Concretely, we introduce (i) a conserved matter current J^μ that determines flux through a detector world-tube Σ, and (ii) an effective conserved current $j^\mu_{\rm ev}$ whose integral counts localized detector-relevant events and thereby defines a discrete clock. A controlled coarse-graining over a scale ℓ produces an infrared clock field $T(x)$, which plays the role of a continuum time variable on scales $\gg \ell$.

The payoff is twofold. First, the TOA observable is manifestly covariant and does not require postulating a self-adjoint time operator. Second, the clock sector carries dynamics: an effective field theory for T introduces correlation scales that enter arrival-time statistics in a parameter-controlled way.

Relation to existing time-of-arrival formalisms. The TOA literature offers several workable definitions—POVM prescriptions, explicit detector models, and path-integral or histories-based constructions. Our goal is not to compete with their operational success, but to isolate a covariant object that can be stated directly at the field level. The key move is to treat the clock as part of the physics: TOA is defined by conditioning flux through Σ on a clock reading built from an event current. In that sense, "detector independence" is not achieved by abstracting away measurement, but by specifying the minimal conserved structures that any detector model must reduce to in the appropriate limit.

Relation to Page–Wootters and relational-clock frameworks. The present construction is complementary to Page–Wootters-type and relational-clock frameworks rather than a replacement for them. In those approaches, time is typically recovered through conditional states or correlations between a clock subsystem and the rest of the system. Here, by contrast, the emphasis is not on Hilbert-space factorization or conditional amplitudes, but on a covariant field-level observable: the detector flux through a world-tube Σ conditioned on a local clock readout generated by an effective event current. The novelty of the present formulation is therefore structural rather than metaphysical: it packages relational time-of-arrival directly as a covariant detector functional with explicit UV-to-IR matching between discrete event counts and a continuum clock field, together with a calculable broadening formula in the clock-EFT regime. In this sense the construction is in dialogue with related perspectives on relational clocks, quantum reference frames, and operational notions of time [12, 13, 14], while keeping the main object of interest at the level of a covariant detector functional rather than a conditional state construction.

Rather than revising measurement postulates or postulating a generalized time operator, we keep the standard field-theoretic setting and change only what is taken to be primitive: TOA is defined as a relational field-functional built from conserved currents and a physical clock sector.

The absence of a universal time operator is then not a gap to be patched, but a cue that TOA should be formulated relationally through correlations between physical degrees of freedom.

Scope and assumptions. Throughout this work, the event current $j^\mu_{\rm ev}$ is treated as a *physical but model-agnostic* structure. It may be fundamental (e.g. a primitive detection or interaction current) or emergent (e.g. a coarse-grained description of localized spacetime processes such as interaction vertices, defect crossings, or excitation worldlines). The approximate conservation $\nabla_\mu j^\mu_{\rm ev} \approx 0$ is expected to hold whenever the relevant events are robust under smooth deformations on the scales of interest, with violations suppressed by the ultraviolet resolution scale.

The coarse-graining map $(\cdot)_\ell$ is assumed to be a local, causal smoothing operation over spacetime regions of characteristic size ℓ, preserving current conservation in the infrared. No specific kernel is fixed; admissible choices include covariant averaging, block-spin–type maps, or detector-induced temporal binning, provided they respect locality and current conservation at scales $\gg \ell$. Admissibility criteria for $j^\mu_{\rm ev}$ in a given experimental context are discussed further in Section 10.

2.2 Detector World-Tubes and Matter Flux

We work on a four-dimensional Lorentzian spacetime $(\mathcal{M}, g_{\mu\nu})$. A detector is modeled as a timelike world-tube $\Sigma \subset \mathcal{M}$, with intrinsic coordinates σ^a $(a = 1, 2, 3)$, induced metric h_{ab}, and future-directed unit normal n_μ.

Let J^μ denote a conserved matter current,

$$\nabla_\mu J^\mu = 0. \tag{2.2.1}$$

Definition 2.2.1 (Matter flux)**.** *The matter flux through Σ is*

$$(\sigma) \equiv n_\mu J^\mu(X(\sigma)). \tag{2.2.2}$$

The total registered flux is

$$\mathcal{N} = \int_\Sigma \mathrm{d}^3\sigma \, \sqrt{|h|} \, (\sigma). \tag{2.2.3}$$

2.3 Event Currents and Discrete Clocks

Definition 2.3.1 (Effective event current)**.** *An* effective event current *is a vector field $j^\mu_{\rm ev}$ satisfying*

$$\nabla_\mu j^\mu_{\rm ev} \approx 0, \tag{2.3.1}$$

where equality may be exact or hold up to controlled corrections on the scales of interest.

Remark 2.3.1 (Status of the event current)**.** *The current $j^\mu_{\rm ev}$ is not assumed to be unique or fundamental. Rather, it represents a coarse-grained, detector-relevant counting current associated with a specified measurement context. Different admissible choices of $j^\mu_{\rm ev}$ correspond, in general, to different effective clock theories and therefore to different TOA statistics. This freedom is not hidden in the formalism: it is part of the model specification and is, in principle, experimentally distinguishable through the resulting clock correlators and arrival-time distributions.*

Remark 2.3.2 (Concrete example)**.** *In a relativistic QFT scattering experiment, $j^\mu_{\rm ev}$ may represent the effective flux of detector-registered interaction events, i.e. a spacetime current whose integral counts localized vertex/crossing events relevant to the measurement.*

Definition 2.3.2 (Event count)**.** *The associated event count through a hypersurface Σ is*

$$N_{\rm ev}(\Sigma) \equiv \int_\Sigma \mathrm{d}\Sigma_\mu \, j^\mu_{\rm ev}. \tag{2.3.2}$$

Proposition 2.3.1. *If $\nabla_\mu j^\mu_{\rm ev} = 0$, then $N_{\rm ev}(\Sigma)$ is invariant under smooth deformations of Σ that preserve its boundary.*

Sketch. The difference between the integrals over two such hypersurfaces is the flux of $\nabla_\mu j^\mu_{\rm ev}$ through the interpolating spacetime region. The claim then follows from Stokes' theorem and current conservation. $\square$

The discrete-event clock is

$$T_{\rm disc} \equiv N_{\rm ev}. \tag{2.3.3}$$

Conceptual hierarchy. The logical structure of the construction may be summarized schematically as

$$\begin{array}{ccc} \text{event current } j^\mu_{\rm ev} & \longrightarrow & \text{discrete clock } T_{\rm disc} = N_{\rm ev} \\ & & \downarrow \text{coarse-graining} \\ & & \text{continuum clock field } T(x) \end{array}$$

The event current is the fundamental object; discrete and continuous clocks are its ultraviolet and infrared descriptions, respectively.

2.4 Coarse-Graining and the Continuum Clock Limit

Let $(\cdot)_\ell$ denote coarse-graining over a spacetime scale ℓ. We assume that the coarse-grained event current is locally integrable in the sense that

$$\partial_{[\mu} (j^{\rm ev}_{\nu]})_\ell \approx 0 \tag{2.4.1}$$

on the infrared scales of interest. Under this condition, one may define a local scalar clock field $T(x)$ by

$$\partial_\mu T(x) \equiv \frac{1}{\nu_0} \big(j_{{\rm ev},\mu}(x)\big)_\ell, \tag{2.4.2}$$

where ν_0 has dimensions of events per unit time.

Lemma 2.4.1. *If $\nabla_\mu j^\mu_{\rm ev} = 0$ and the local integrability condition* (2.4.1) *holds after coarse-graining, then $T(x)$ is locally well-defined up to an additive constant; global obstructions correspond to nontrivial topology.*

Proposition 2.4.1 (IR equivalence)**.** *At scales $\gg \ell$, event counts and clock differences match:*

$$N_{\rm ev}(\Sigma) \approx \nu_0 \int_\Sigma {\rm d}\Sigma_\mu\, \partial^\mu T = \nu_0\, \Delta T, \tag{2.4.3}$$

where the last equality holds when Σ interpolates between hypersurfaces of constant T. This is the relevant situation when the detector region is used to compare two clock slices—for example, in the local detector setups considered below, where the clock field is sampled between successive level sets of the coarse-grained readout. For generic finite detectors in $3+1$ dimensions this exact identification need not hold globally, and the equality should then be understood as a local or idealized limit; extending the construction to fully finite detector geometries is part of the program outlined in the Conclusion.

2.5 Clock-Sector Effective Field Theory

For the purpose of a local EFT expansion in a Minkowski background, we choose an inertial coordinate patch and expand the clock field around a monotonically increasing background solution,

$$T(x) = T_0(x) + \tau(x), \qquad T_0(x) = t, \tag{2.5.1}$$

so that τ describes fluctuations around the reference clock profile.

A minimal stable quadratic EFT is

$$S_\tau = \frac{1}{2} \int \mathrm{d}^4x \left[(\partial_t \tau)^2 - c_\tau^2 (\nabla \tau)^2 - \mu_\tau^2 \tau^2 \right]. \tag{2.5.2}$$

The parameter μ_τ sets a correlation scale in space and time.

In Appendix .1, μ_τ is shown to control the exponential suppression envelope of spacelike clock correlations and the corresponding early-arrival bounds.

Remark 2.5.1 (Correlation scale)**.** *The mass parameter μ_τ sets the characteristic temporal and spatial correlation scale of the clock field. As shown explicitly in Appendix .1, the same parameter controls the exponential suppression envelope of spacelike clock correlations and thereby motivates corresponding bounds on early-arrival contributions in TOA models where such contributions are clock-correlation controlled.*

2.6 Relational Time-of-Arrival Observable

Definition 2.6.1 (Relational TOA distribution)**.** *The TOA probability density with respect to the clock field T is*

$$p(\Theta) \equiv \frac{1}{\mathcal{N}} \int_\Sigma \mathrm{d}^3\sigma \sqrt{|h|}\,(\sigma)\, \delta(T_\Sigma(\sigma) - \Theta), \tag{2.6.1}$$

where $T_\Sigma(\sigma) \equiv T(X(\sigma))$.

Why this is nontrivial. Equation (2.6.1) is not a relabeling of an external time coordinate. The clock $T(x)$ is a dynamical field (or effective field) with its own correlators, fixed by the clock-sector EFT. Those correlators enter TOA statistics through the averaging in Eq. (2.8.9) and lead to quantitative modifications—e.g. broadened arrival-time distributions and an μ_τ-controlled envelope for contributions that would require spacelike clock correlations (Appendix .1).

This definition is covariant and compresses detector dependence into the geometry of Σ, the effective flux functional , and the clock-sector correlators.

2.7 Semiclassical Limit

In the semiclassical regime of sharply peaked wavepackets and weak clock fluctuations,

$$p(\Theta) \to \delta(\Theta - \Theta_{\mathrm{cl}}), \tag{2.7.1}$$

where Θ_{cl} matches the classical time-of-flight.

2.8 Worked Example in $1+1$ Dimensions

We now give an explicit $1+1$D computation (wavepacket + clock correlator). This section is intended as a concrete template; the same steps extend to $3+1$D with transverse integrals.

2.8.1 Setup: detector worldline and matter wavepacket

Work in Minkowski space with coordinates (t, x) and metric $\eta = \mathrm{diag}(1, -1)$. Model the detector as the timelike worldline at fixed $x = x_D$:

$$\Sigma : \quad X^\mu(\sigma) = (t = \sigma,\ x = x_D). \tag{2.8.1}$$

For a $1+1$D detector "world-tube", the induced measure reduces to $\mathrm{d}\sigma = \mathrm{d}t$.

Take a complex Klein–Gordon field ψ of mass m with conserved $U(1)$ current [6, 7]:

$$J^\mu = \frac{i}{2m}\left(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*\right), \qquad \partial_\mu J^\mu = 0. \tag{2.8.2}$$

In $1+1$D, the spatial flux at the detector is $J^x(t, x_D)$. We take the incoming direction to be $+x$ and define the registered flux by

$$(t) \equiv J^x(t, x_D)\, 1_{\{J^x \geq 0\}}. \tag{2.8.3}$$

This positive-flux restriction is an operational assumption, standard in TOA constructions [2, 3, 4], and is appropriate in the narrowband, predominantly right-moving regime considered below. In that regime the negative-flux component is parametrically suppressed, so Eq. (2.8.3) modifies neither the normalization nor the convolution formula beyond subleading corrections. Outside that regime, backflow or bidirectional contributions must be treated explicitly and the simple positive-flux truncation need not be adequate.

Let the positive-frequency wavepacket be

$$\psi(t, x) = \int_{-\infty}^{+\infty} \frac{\mathrm{d}k}{\sqrt{2\pi}}\, a(k)\, e^{-i\omega_k t + ikx}, \qquad \omega_k = \sqrt{k^2 + m^2}. \tag{2.8.4}$$

Choose $a(k)$ narrowly peaked at $k_0 > 0$ with width $\Delta k \ll k_0$. Then the packet propagates with group velocity

$$v_g = \frac{\partial \omega_k}{\partial k}\Big|_{k_0} = \frac{k_0}{\omega_{k_0}} \in (0, 1). \tag{2.8.5}$$

2.8.2 Semiclassical approximation for the flux

Under the narrowband approximation, J^x reduces to the familiar form

$$J^x(t, x) \approx v_g\, |\psi(t, x)|^2, \tag{2.8.6}$$

up to corrections of order $\mathcal{O}((\Delta k/k_0)^2)$. A Gaussian $a(k)$ gives an approximately Gaussian arrival profile at $x = x_D$:

$$(t) \approx_0 \exp\left[-\frac{(t - t_{\mathrm{cl}})^2}{2\sigma_t^2}\right], \qquad t_{\mathrm{cl}} = \frac{x_D - x_0}{v_g}, \tag{2.8.7}$$

where x_0 is the initial packet center and $\sigma_t \sim 1/(v_g \Delta k)$. Normalize $\mathcal{N} = \int \mathrm{d}t\, (t)$.

2.8.3 Clock field on the detector and stationary correlator

Take the continuum clock field

$$T(t, x) = t + \tau(t, x). \tag{2.8.8}$$

The relational TOA density (2.6.1) becomes in $1+1$D

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_{-\infty}^{+\infty} \mathrm{d}t\, (t) \left\langle \delta\big(t + \tau(t, x_D) - \Theta\big) \right\rangle_\tau, \tag{2.8.9}$$

where $\langle\cdot\rangle_\tau$ denotes averaging over clock fluctuations.

Assume the clock sector is in a stationary Gaussian state induced by the quadratic EFT (2.5.2). Then the single-point random variable $\tau_\ell(t, x_D)$ is Gaussian, with mean 0 and variance

$$\sigma^2_{\tau,\ell} \equiv \langle \tau_\ell(t, x_D)^2 \rangle = C_\ell(0, 0), \tag{2.8.10}$$

where the two-point function is

$$C(\Delta t, \Delta x) \equiv \langle \tau(t, x)\tau(t + \Delta t, x + \Delta x)\rangle. \tag{2.8.11}$$

Here τ_ℓ denotes the clock field smeared over the detector's finite spacetime resolution scale ℓ. This coarse-grained variance is the physically relevant quantity entering the TOA distribution. In a continuum free-field description, the unsmeared coincidence limit $C(0, 0)$ is UV-sensitive in $1+1$ dimensions, so the theory should be understood as defined either with an explicit ultraviolet cutoff Λ or, equivalently for present purposes, with a finite detector/coarse-graining scale ℓ. Accordingly, the observable variance is really $\sigma^2_{\tau,\ell} = \sigma^2_{\tau,\ell}(\mu_\tau)$, or equivalently $\sigma^2_\tau(\Lambda, \mu_\tau)$ in cutoff language.

In $1 + 1$D, for a gapped relativistic scalar ($c_\tau = 1$) at equal position $\Delta x = 0$, the stationary correlator along the detector worldline decays on the characteristic timescale μ_τ^{-1}. The timelike continuation of the massive correlator yields an oscillatory Bessel/Hankel-type envelope with the same large-$|\Delta t|$ decay scale, so that

$$C(\Delta t, 0) \sim (\text{oscillatory Bessel/Hankel prefactor}) \times e^{-\mu_\tau |\Delta t|} \quad \text{for} \quad |\Delta t| \gg \mu_\tau^{-1}, \tag{2.8.12}$$

up to representation-dependent prefactors and analytic continuation conventions. Appendix .1 derives the corresponding exponential envelope explicitly for spacelike separations; the present timelike statement is the detector-worldline analogue and is used here only at the level of its characteristic decay scale.

2.8.4 Closed-form TOA distribution: convolution formula

For a stationary Gaussian variable $X = \tau_\ell(t, x_D)$ with variance $\sigma^2_{\tau,\ell}$,

$$\langle \delta(t + X - \Theta)\rangle_X = \frac{1}{\sqrt{2\pi\sigma^2_{\tau,\ell}}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma^2_{\tau,\ell}}\right]. \tag{2.8.13}$$

To obtain a closed convolution form, we further assume that, at leading order, the local clock fluctuation $\tau(t, x_D)$ is statistically independent of the matter-flux profile $\mathcal{F}(t)$. Physically, this corresponds to a weak-coupling or spectator-clock regime in which the clock sector broadens the readout without appreciably backreacting on the matter current. If clock–matter correlations are appreciable, mixed correlators contribute and the simple convolution formula below is replaced by a more general conditional-average expression.

Under the Gaussian and leading-order independence assumptions just stated, Eq. (2.8.9) becomes the convolution

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_{-\infty}^{+\infty} \mathrm{d}t\, (t)\, \frac{1}{\sqrt{2\pi\sigma^2_{\tau,\ell}}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma^2_{\tau,\ell}}\right]. \tag{2.8.14}$$

If (t) is Gaussian as in (2.8.7), the convolution is again Gaussian:

$$p(\Theta) = \frac{1}{\sqrt{2\pi(\sigma_t^2 + \sigma^2_{\tau,\ell})}} \exp\left[-\frac{(\Theta - t_{\mathrm{cl}})^2}{2(\sigma_t^2 + \sigma^2_{\tau,\ell})}\right]. \tag{2.8.15}$$

Thus the clock sector produces a *predictive broadening*:

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma^2_{\tau,\ell}, \qquad \sigma^2_{\tau,\ell} = C_\ell(0, 0). \tag{2.8.16}$$

Falsifiability. Given an experimentally or numerically specified flux profile $\mathcal{F}(t)$ and a measured or modeled clock variance $\sigma^2_{\tau,\ell}$, the convolution formula (2.8.14) predicts a unique TOA distribution without additional fitting parameters. Deviations from the predicted variance $\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma^2_{\tau,\ell}$ or from the expected crossover behavior at $\Delta t \sim \mu_\tau^{-1}$ would directly falsify the assumed clock sector or its effective field description.

Remark 2.8.1 (Gaussian tails vs exponential envelopes)**.** *The Gaussian form of* $p(\Theta)$ *arises from the* local, one-point *statistics of the clock readout at the detector. Exponential behavior enters instead through the* spacetime propagation *of clock correlations: early-arrival contributions requiring spacelike correlations are bounded by an exponential* envelope *set by* μ_τ^{-1}*, derived in Appendix .1. These two effects are conceptually distinct and govern different regimes of the TOA distribution.*

2.8.5 Discrete-event ultraviolet corrections

If the UV clock is an event-count $T_{\mathrm{disc}} = N_{\mathrm{ev}}$, then at time resolution $\Delta t \lesssim \ell$ one expects renewal-type corrections:

$$p(\Theta) \;=\; p_{\mathrm{IR}}(\Theta) \;+\; \delta p_{\mathrm{UV}}(\Theta), \tag{2.8.17}$$

where p_{IR} is the continuum prediction (e.g. (2.8.15)) and δp_{UV} carries integer-step features and non-Gaussianity that vanish upon coarse-graining $\Delta t \gg \ell$.

Bridge to a concrete realization. So far the discussion has been model-independent: J^μ and j^μ_{ev} are treated as effective conserved structures. The next section simply records one explicit realization in which j^μ_{ev} is generated by defect worldlines; none of the preceding definitions rely on that choice.

2.9 Example Microscopic Realization

The relational TOA construction above is model-independent: it requires only (i) a conserved matter flux J^μ through the detector world-tube Σ and (ii) a conserved (or approximately conserved) event current j^μ_{ev} whose integral defines the ultraviolet event-count clock. Here we record one concrete realization in a hydrodynamic/defect setting motivated by Swirl-String Theory (SST), presented strictly as an example.

2.9.1 Defect (knot-center) worldline current

Consider N stable, localized defects with center worldlines $z^\mu_k(\tau)$. Define the distributional defect current

$$j^\mu_{\mathrm{ev}}(x) \;\equiv\; \sum_{k=1}^{N} q_k \int \mathrm{d}\tau\, \frac{\mathrm{d}z^\mu_k}{\mathrm{d}\tau}\, \delta^{(4)}\big(x - z_k(\tau)\big), \tag{2.9.1}$$

where $q_k \in \mathbb{Z}$ is an integer-valued defect charge. In the absence of creation/annihilation events in the bulk, this current is conserved in the distributional sense,

$$\nabla_\mu j^\mu_{\mathrm{ev}}(x) = 0, \tag{2.9.2}$$

and the associated event count through a detector world-tube Σ is

$$N_{\mathrm{ev}}(\Sigma) \equiv \int_\Sigma \mathrm{d}\Sigma_\mu\, j^\mu_{\mathrm{ev}}. \tag{2.9.3}$$

Operationally, $N_{\mathrm{ev}}(\Sigma)$ counts (with multiplicity q_k) the number of defect worldlines intersecting the detector region.

2.9.2 Continuum clock field from coarse-graining

At scales large compared to a coarse-graining length ℓ, define a smooth clock field $\chi(x)$ by

$$\partial_\mu \chi(x) \equiv \frac{1}{\nu_0} \left(j_{\mathrm{ev},\mu}(x) \right)_\ell, \tag{2.9.4}$$

where $(\cdot)_\ell$ denotes coarse-graining and ν_0 is a conversion factor. This yields the continuum-clock (IR) description of the event-count (UV) clock.

2.9.3 Foliation interpretation

If the macroscopic medium admits a hypersurface-orthogonal timelike congruence u_μ (a foliation), and if $\nabla_\mu \chi$ is everywhere timelike in the region of interest, then one may parametrize it by a scalar potential χ via

$$u_\mu = \frac{\nabla_\mu \chi}{\sqrt{g^{\alpha\beta} \nabla_\alpha \chi \, \nabla_\beta \chi}}. \tag{2.9.5}$$

In this interpretation, the TOA observable $p(\Theta)$ defined in Eq. (2.6.1) becomes the distribution of detector flux binned by the physical clock reading χ on Σ.

2.9.4 Interpretational note (optional)

In SST language one may view the localized defects as "topological matter excitations" in the effective medium. The integer charges q_k can then be interpreted as quantized defect labels. Classical hydrodynamic analogies (e.g., linking/helicity-based interpretations) are deferred to Discussion to avoid placing additional structure into the formal construction.

2.10 Discussion and Outlook (Analogy Only)

What the event current represents (and what it does not). In the main text, the event current j^μ_{ev} is introduced as an (approximately) conserved quantity whose integral counts *localized, robust events* relevant to a detection process. At this level of generality it is deliberately agnostic: the "events" might be particle crossings of a detector region, interaction vertices effectively registered by an apparatus, defect/excitation worldlines intersecting Σ, or other countable processes whose total number is stable under smooth deformations of Σ that keep the boundary fixed.

This is the only structural assumption the formalism needs: that on the scales of interest the relevant events can be summarized by a current with $\nabla_\mu j^\mu_{\mathrm{ev}} \approx 0$ (exactly or with controlled violations). Section 2.9 provides one concrete realization in terms of defect worldlines in a hydrodynamic medium, but the relational TOA definition itself does not depend on adopting that ontology. Put differently, the construction is compatible with many microscopic stories; it only requires that they reduce to an effective event-count description on appropriate scales.

Classical analogy (optional intuition, not an assumption). The comparison with helicity in ideal fluid dynamics is meant as a guide to intuition rather than a derivation. In fluids, conserved integrals can often be understood geometrically in terms of linking and knotting of flow structures [9, 10, 11]. By analogy, one may picture j^μ_{ev} as a flux of "crossing events" that are topologically protected or otherwise robust, so that a discrete count becomes a smooth clock field under coarse-graining. None of the results above require this picture; it merely suggests one way in which a conserved event current might arise in an effective description.

Predictive content and where it enters. Once a clock sector is specified, its correlators $C(\Delta t, \Delta x)$ become physical input rather than interpretational decoration. In the simplest

(Gaussian) setting they determine the amount of arrival-time broadening through local clock variance, while massive clock correlations also supply an exponential envelope for contributions that would require correlations across a spacelike separation (Appendix .1). At sufficiently fine time resolution, the ultraviolet event-count description can additionally produce discrete/renewal features that are washed out in the infrared after coarse-graining.

Admissibility of event currents. The present formalism does not claim that every approximately conserved current is equally natural in a given experiment. The selection of an admissible $j^{\mu}_{\rm ev}$ is part of the effective model-building step and should be constrained by the detector context: locality, robustness under coarse-graining, compatibility with the detector's counting statistics, and empirical stability across changes of resolution. In this sense, the framework is not falsified or confirmed by a current in the abstract, but by whether a specified admissible current yields a clock sector whose correlators and TOA statistics agree with observation. A systematic classification of admissible event currents is left for future work.

Idealized measurement scenarios

A minimal scenario is a relativistic wavepacket experiment in which the detector couples (directly or effectively) to a clock degree of freedom with a finite correlation time. The observable consequence is that the TOA statistics need not coincide with those obtained from the matter wavepacket alone. In the Gaussian clock limit discussed in Sec. 8, the predicted broadening

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma_{\tau,\ell}^2 \tag{2.10.1}$$

would appear as an excess arrival-time spread beyond the semiclassical flux width σ_t.

From a practical standpoint, the framework is well-suited to numerical study. Given a flux profile $\mathcal{F}(t)$ and a model (or measurement-informed ansatz) for the clock correlator $C(\Delta t, \Delta x)$, the TOA density follows directly from the convolution formula, Eq. (2.8.14). This makes the clock-sector parameters (e.g. c_τ and μ_τ in the quadratic EFT) quantitatively testable in toy models and in lattice or continuum simulations.

2.11 Conclusion

We defined time-of-arrival (TOA) as a *relational*, manifestly covariant observable: detector crossings are binned neither by an externally supplied time coordinate nor by postulating a universal time operator, but by a *physical clock* evaluated on the detector world-tube Σ, within the class of clocks that admit a (conserved or approximately conserved) event current. The construction uses two ingredients:

(i) a conserved matter current J^μ whose contraction with the unit normal defines the detector flux $\mathcal{F} = n_\mu J^\mu$, and

(ii) an event current $j^{\mu}_{\rm ev}$ whose integral defines a discrete event-count clock. The choice of $j^{\mu}_{\rm ev}$ is part of the effective model specification rather than an unobservable convention: different admissible event currents define different clock sectors and are, in principle, empirically distinguishable through the resulting TOA statistics.

A controlled coarse-graining over a scale ℓ then yields an infrared clock field $T(x)$ through

$$\partial_\mu T(x) = \frac{1}{\nu_0} \big(j_{{\rm ev},\mu}(x)\big)_\ell, \tag{2.11.1}$$

so that discrete counts and continuum time arise as UV and IR descriptions of the same underlying structure.

The resulting TOA density,

$$p(\Theta) = \frac{1}{\mathcal{N}} \int_\Sigma \mathrm{d}^3\sigma \, \sqrt{|h|}\, \mathcal{F}(\sigma)\, \delta(T_\Sigma(\sigma) - \Theta), \tag{2.11.2}$$

depends on detector microphysics only through the geometry of Σ, the effective flux J^μ, and the clock-sector correlators. Because the definition never requires a self-adjoint operator conjugate to the Hamiltonian, it sidesteps Pauli-type obstructions at the outset. In the semiclassical regime it reproduces the usual time-of-flight notion.

In the worked $1+1$D example, a narrowband incoming wavepacket together with a stationary Gaussian clock sector reduces $p(\Theta)$ to a convolution of the semiclassical flux profile with the local clock readout distribution, producing a concrete prediction: arrival-time broadening

$$\mathrm{Var}(\Theta) = \sigma_t^2 + \sigma_{\tau,\ell}^2, \qquad \sigma_{\tau,\ell}^2 = \langle \tau_\ell^2 \rangle, \tag{2.11.3}$$

with $\sigma_{\tau,\ell}^2$ fixed by the clock sector together with the detector/coarse-graining scale. Beyond this local effect, Appendix .1 shows that massive clock correlators are exponentially suppressed at spacelike separations with envelope scale μ_τ^{-1} (up to algebraic prefactors), providing a controlled bound on early-arrival contributions without implying superluminal signaling.

Relative to mainstream TOA formalisms, the point here is not that detector models or POVMs fail operationally, but that the relational definition isolates a covariant field-functional whose dependence on measurement details is pushed into explicit, modelable structures: conserved currents and a clock sector with its own dynamics. This reframes detector dependence as a UV→IR matching problem—event counts to continuum time—making TOA a covariant, predictive field observable.

Next steps

Immediate extensions include: (i) a $3+1$D worked example for finite-size detectors to quantify geometric dependence; (ii) non-Gaussian clock dynamics beyond the quadratic EFT to predict higher-cumulant signatures in $p(\Theta)$; and (iii) a systematic treatment of renewal-type UV corrections at temporal resolution $\Delta t \lesssim \ell$ and their crossover to the continuum limit. In particular, a numerical study of Eq. (2.8.14) as μ_τ is varied would directly probe the predicted exponential suppression envelope and the crossover from discrete event-count statistics to continuum clock time.

Appendix

.1 Exponential Suppression from the Clock Propagator

This appendix derives the exponential suppression scale governing spacelike clock correlations and motivates the corresponding exponential envelope expected in early-arrival tails for models in which such tails are controlled by clock-sector correlations. The derivation uses standard massive-scalar correlators and Bessel function asymptotics [6, 7, 8].

.1.1 Clock correlators from the quadratic EFT

Consider the quadratic clock action in flat $1+1$D with $c_\tau = 1$:

$$S_\tau = \frac{1}{2}\int \mathrm{d}^2x \left[(\partial_t\tau)^2 - (\partial_x\tau)^2 - \mu_\tau^2\tau^2\right]. \tag{.1.1}$$

The Wightman function $G^+(x)$ for a free massive scalar is a Lorentz-invariant function of $s^2 = (\Delta t)^2 - (\Delta x)^2$:

$$G^+(\Delta t, \Delta x) = \langle \tau(t,x)\tau(t+\Delta t, x+\Delta x)\rangle. \tag{.1.2}$$

For spacelike separations $(\Delta x)^2 - (\Delta t)^2 > 0$, standard results give (see, e.g., representations in [7] and integral tables):

$$G^+(\Delta t, \Delta x) = \frac{1}{2\pi}K_0\Big(\mu_\tau\sqrt{(\Delta x)^2 - (\Delta t)^2}\Big) \quad \text{(spacelike)}. \tag{.1.3}$$

While conventions vary by factors and $i0^+$ prescriptions, the key fact is that the spacelike decay is controlled by K_0.

.1.2 Bessel asymptotics and exponential decay

For large argument $z \gg 1$, the modified Bessel function satisfies

$$K_0(z) \sim \sqrt{\frac{\pi}{2z}}\, e^{-z} \left[1 + \mathcal{O}\!\left(\frac{1}{z}\right)\right]. \tag{.1.4}$$

Therefore, for spacelike separations with invariant distance

$$\rho \equiv \sqrt{(\Delta x)^2 - (\Delta t)^2} \gg \mu_\tau^{-1}, \tag{.1.5}$$

Eq. (.1.3) yields the exponential suppression

$$G^+(\Delta t, \Delta x) \;\propto\; \frac{e^{-\mu_\tau \rho}}{\sqrt{\mu_\tau \rho}} \left[1 + \mathcal{O}\Big((\mu_\tau \rho)^{-1}\Big)\right]. \tag{.1.6}$$

Proposition .1.1 (Spacelike correlator envelope)**.** *For the quadratic clock EFT, the spacelike two-point function is exponentially suppressed with characteristic envelope scale μ_τ^{-1} as in* (.1.6)*.*

Remark .1.1. *This mechanism is directly analogous to standard QFT statements: massive fields do not transmit strong correlations outside the light cone; the residual correlations decay as $e^{-\mu\rho}$ (up to power-law prefactors). This is a correlator statement and does not imply superluminal signaling.*

.1.3 Heuristic implication for early-arrival tails

We now state a physically motivated inference rather than a theorem. In any TOA model in which contributions to sufficiently early detector readout require clock-sector correlations across an effectively spacelike interval, the spacelike suppression (.1.6) suggests an exponential envelope of the form

$$p_{\text{early}}(\Delta) \;\lesssim\; A(\Delta)\, \exp\big(-\mu_\tau\, \rho(\Delta)\big), \tag{.1.7}$$

where Δ parameterizes the degree of earliness and $\rho(\Delta)$ denotes the corresponding spacelike separation scale.

Here $A(\Delta)$ is left unspecified; in concrete realizations it must be derived from the detailed coupling between the clock sector and the TOA statistic. Accordingly, Eq. (.1.7) should be read as a heuristic envelope statement induced by the clock correlator, not as a model-independent theorem for all TOA constructions.

.1.4 Retarded Green's function and locality

For completeness, the retarded propagator solves

$$(\Box + \mu_\tau^2)\, G_{\text{ret}}(x) = \delta^{(2)}(x), \qquad G_{\text{ret}}(x) = 0 \text{ for } t < 0. \tag{.1.8}$$

Local detector couplings to τ then inherit causal support from G_{ret}, while spacelike Wightman correlations remain exponentially suppressed as above. This distinguishes *correlation tails* from *signal propagation*, consistent with standard relativistic QFT [6, 7].

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All data generated or analyzed during this study are included in this published article.

Code availability

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Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

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Appendix A

SST-50 — Emergent equivalence principle

[theorem]Proposition
Remark
Definition
[Relational Time-of-Arrival]Time from Swirl: A Hydrodynamic Origin of Proper Time
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Abstract

The Equivalence Principle (EP) has long anchored the geometric reading of gravitation, yet its foundational status is less clear than the historical narrative suggests. Three lines of work point in the same direction: (i) relativistic kinematics can be fixed on operational grounds from causality alone; (ii) relational formulations of quantum dynamics generate effective time-evolution from clock–system correlations; and (iii) the "geometric trinity" shows that curvature-, torsion-, and non-metricity-based formulations are dynamically equivalent while conceptually distinct. I bring these strands together in a conservative framework. Lorentzian causal structure is treated as operationally primary. Mild deformations of a relational Hamiltonian constraint yield gravitational redshift and Newtonian-like interaction terms at low energy. In this setting, the EP reads most naturally as an *emergent* organizing symmetry rather than a primitive axiom. I isolate which parts of the usual EP story are logically prior, which are recovered asymptotically, and where coherence-dependent corrections to redshift should appear. The role of the affine connection is made explicit in mediating inertial and gravitational effects, and the operational regime of validity is expressed through dimensionless parameters directly measurable in quantum-clock interferometry. relational time, time of arrival, quantum time observables, covariant observables, event currents, quantum field theory

A.1 Introduction

The Equivalence Principle (EP) is often presented as the conceptual cornerstone of General Relativity (GR), justifying the identification of gravity with spacetime geometry and the universality of free fall [17]. At the same time, several mature ideas now suggest a different ordering of concepts. First, special-relativistic kinematics can be derived from assumptions about observers, signals, and causal order, with no privileged role for electromagnetism [8, 9]. Second, in relational-time approaches to quantum dynamics, effective redshift and Newtonian-like terms arise from clock–energy couplings in a global constraint, even for otherwise non-interacting subsystems [20, 11]. Third, the geometric trinity shows that curvature-, torsion-, and non-metricity-based theories can be dynamically equivalent while remaining conceptually distinct [14, 16, 15].

Aim. I give a structural account that respects established phenomenology while reordering the logical dependencies: (i) causality fixes the inertial scaffold; (ii) relational time generates effective gravitational structure at low energy; and (iii) the same empirical content admits inequivalent geometric codings. Read this way, the EP becomes an emergent low-energy symmetry.

Scope. No new long-range forces are proposed, and no composition-dependent violations are assumed. The goal is to clarify the domain in which the EP is best interpreted as emergent and to identify concrete, coherence-sensitive signatures for quantum-clock experiments.

Contributions.

1. An operational summary of how causal order fixes Lorentzian inertial structure (Sec. A.2).
2. A compact operator statement for the low-energy expansion of the relational effective Hamiltonian and its universal energy–energy cross-term (Sec. A.3, Prop. A.3.1).
3. A perspectival reading that reconciles relational time with connection-based encodings without a globally preferred clock (Sec. A.4).
4. A clear account of weak-field redshift and the Newtonian limit in the relational picture, and what is still open about distance dependence (Sec. A.5).
5. A use of the geometric trinity as geometric underdetermination of shared empirical content (Sec. A.6).
6. A synthesis in which the EP is an emergent organizing symmetry, with quantum-clock tests singled out as near-term probes (Secs. A.8–A.9).

Remark A.1.1 (Didactic recap)**.** In one paragraph. *The paper keeps GR phenomenology intact but reorders the logic: causality fixes the Lorentzian scaffold; relational time with mild constraint deformations generates redshift and a universal energy–energy coupling; and the geometric trinity shows multiple, conceptually distinct encodings with the same dynamics. In that regime, the EP reads as a low-energy symmetry, not a primitive axiom.*

A.2 Causality as the Origin of Relativistic Kinematics

A.2.1 Observers, events, and an operational start

Treat an observer as a localized system carrying a clock and the ability to send/receive signals. Events are indexed by that clock's reading. This minimal setup captures the idea that spacetime notions are fixed by procedures of registration and communication [8].

A.2.2 Finite signaling and a universal distance

If exchanging information takes finite time, round-trip signal times define distance. Minimizing over admissible messengers picks out a limiting signal class and a maximal transfer speed, with no need to reference a specific field [8].

A.2.3 Causal automorphisms and the Lorentz group

Under mild spatial regularity (Euclidean distance in inertial frames), transformations that preserve causal order form the inhomogeneous Lorentz group (up to dilatations) by Alexandrov–Zeeman-type results [9]. Causality thus fixes the Lorentzian kinematic backbone.

Remark A.2.1 (Didactic recap)**.** Takeaway. *Start from clocks and signals, not fields. Finite signaling ⇒ operational distance; preserving causal order ⇒ Lorentz group (modulo scale). This supplies the inertial structure used later by the relational-time construction.*

A.3 Relational Time in Quantum Mechanics

A.3.1 Page–Wootters in brief

Relational-time schemes encode dynamics in correlations between a clock and a system. A stationary global constraint coexists with conditional Schrödinger evolution at fixed clock readings [11, 12, 13].

A.3.2 A simple deformation and clock–energy coupling

To model backreaction-like effects, consider

$$\hat{J} \equiv \hat{p}_t \otimes \hat{I}_S + \hat{I}_t \otimes \hat{H}_S + \frac{1}{\Lambda}\,\hat{p}_t \otimes \hat{H}_S \approx 0, \tag{A.3.1}$$

with deformation scale Λ [20]. Conditioning on the clock yields an energy-dependent phase rate summarized by

$$\hat{H}_{\text{eff}} = \hat{H}_S\big(\hat{I}_S + \hat{H}_S/\Lambda\big)^{-1}, \tag{A.3.2}$$

on suitable domains (App. .1).

The parameter Λ plays the role of an inverse deformation scale, with $\varepsilon_\Lambda = \|\hat{H}_S\|/\Lambda \ll 1$ defining the domain of validity for the perturbative expansion. Operationally, Λ^{-1} encodes the strength of backreaction between the system and the global clock, analogous to a weak coupling constant in effective theory.

Proposition A.3.1 (Low-energy expansion of the relational effective Hamiltonian)**.** *Let $\hat{H}_S$ be self-adjoint and spectrally supported on a subspace with $\|\hat{H}_S\| < \Lambda$ for some $\Lambda > 0$. Then*

$$(\hat{I}_S + \hat{H}_S/\Lambda)^{-1} = \sum_{n=0}^{\infty}(-1)^n(\hat{H}_S/\Lambda)^n,$$

so

$$\hat{H}_{\text{eff}} = \hat{H}_S - \frac{1}{\Lambda}\hat{H}_S^2 + O(\Lambda^{-2}) \qquad (\|\hat{H}_S\|/\Lambda \ll 1).$$

If $\hat{H}_S = \hat{H}_A + \hat{H}_B$ with $[\hat{H}_A, \hat{H}_B] = 0$ on that subspace, then the $O(\Lambda^{-1})$ term includes the universal cross-term $-(2/\Lambda)\,\hat{H}_A \otimes \hat{H}_B$.

Remark A.3.1. *The spectral restriction can be read as a low-energy condition or enforced by a bounded-spectrum truncation of the clock/system. That is the regime in which Λ^{-1} is a controlled perturbation [20].*

A.3.3 Composite systems and emergent interaction

For $\hat{H}_S = \hat{H}_A + \hat{H}_B$ with $|\hat{H}_S| \ll \Lambda$, expanding (A.3.2) gives

$$\hat{H}_{\text{eff}} \approx \hat{H}_A + \hat{H}_B - \frac{1}{\Lambda}(\hat{H}_A^2 + \hat{H}_B^2 + 2\,\hat{H}_A \otimes \hat{H}_B) + O(\Lambda^{-2}), \tag{A.3.3}$$

i.e., a universal energy–energy coupling. With an external identification of distance, this matches the Newtonian form in the weak-field limit (Sec. A.5).

Remark A.3.2 (Didactic recap)**.** Plain language. *The clock couples to the system through the constraint. At low energy, that acts like a small correction to the system Hamiltonian: a universal energy–energy term appears between parts A and B. This is the seed of both redshift and Newtonian-like behavior later on.*

A.4 Perspectival Structure of Relational Time and Gravity

A durable lesson from quantum foundations is that physically meaningful descriptions are local to a context. In an endo-theoretic perspectival view, perspectives come with finite resolution and context-dependent Boolean algebras; a single global Boolean valuation is unavailable [21]. Page–Wootters-type constructions fit this picture: each clock choice defines a local temporal ordering while the global state remains stationary [11, 12].

When gravitational effects are modeled by universal clock–energy couplings, proper time is best treated as an observable in correlations. Gravitational time dilation then records a mismatch between clock perspectives rather than the action of a background metric [20]. The geometric trinity is compatible with this: distinct geometric codings (curvature, torsion, non-metricity) summarize the same local dynamical content [16, 15, 14]. In that light, the EP reads as a low-energy compatibility condition among local perspectives.

Remark A.4.1 (Didactic recap)**.** Intuition. *No single, globally preferred clock; each clock defines a valid local time. Geometry (curvature/torsion/non-metricity) is then a way to* summarize *how these local descriptions fit together, not a unique underlying picture.*

A.5 Gravitational Redshift and the Newtonian Limit from Relational Dynamics

A.5.1 Proper time as an internal observable

Model a subsystem as a clock and compare its conditional evolution with a coordinate time derived from the global clock. Proper time is then an observable, not a background parameter [20].

A.5.2 Weak-field redshift

In the low-energy regime, the deformation scale Λ modifies the clock's tick rate relative to coordinate time. Identifying Λ^{-1} with a Newtonian potential scale reproduces leading-order gravitational time dilation [20, 17].

A.5.3 On distance dependence

The relational model does not itself enforce a $1/r$ law unless spatial relations are operationally reconstructed. What emerges directly is a universal energy–energy coupling and a redshift structure; mapping these to a Newtonian potential GM/r requires defining distance through signal-exchange procedures [8]. A consistent approach is to build an operational metric from round-trip times of limiting signals within the causal scaffold, then embed that definition into the relational constraint. This would promote $1/r$ behavior from a phenomenological identification to a derived consequence of the combined causal and relational structure.

A.5.4 High-energy saturation

Because (A.3.2) saturates phase rates at high energy, certain ultraviolet behaviors are softened. I treat this as an effective-theory feature rather than a fundamental claim [20].

Remark A.5.1 (Didactic recap)**.** Checklist. *(1) Redshift emerges from clock–energy coupling; (2) Newtonian form needs a distance model; (3)* Λ *sets the small parameter; (4) the construction softens some UV behavior without claiming a full quantum-gravity completion.*

A.6 Metric–Affine Gravity and the Geometric Trinity

A.6.1 Metric and connection as independent

In metric–affine approaches the metric $g_{\mu\nu}$ and connection $\Gamma^{\lambda}_{\mu\nu}$ are independent. Curvature, torsion, and non-metricity characterize different properties of $\Gamma^{\lambda}_{\mu\nu}$ [15, 16].

A.6.2 GR, TEGR, STEGR: dynamical equivalence

GR (curvature), TEGR (torsion), and STEGR (non-metricity) are dynamically equivalent up to boundary terms, yielding the same classical field equations in their standard forms [16, 15]. The upshot is geometric underdetermination.

A.6.3 Conceptual inequivalence and EP

Despite dynamical equivalence, the part played by the EP differs conceptually, especially regarding the unification of inertial and gravitational effects. See [14] for a detailed mapping.

Remark A.6.1 (Didactic recap)**.** What to remember. *Same classical predictions ≠ same concepts. Curvature, torsion, and non-metricity are different stories that encode the same on-shell content. This is why an* emergent *EP is plausible.*

This triadic structure is summarized visually in Fig. A.1.

[thick, every node/.style=align=center, font=] (Causality) at (90:4) Operational
Causality; (RelationalTime) at (210:4) Relational
Time; (Trinity) at (330:4) Geometric
Trinity;
[-] (Causality) – (RelationalTime) – (Trinity) – (Causality);
[draw, circle, minimum size=2.5cm, inner sep=2pt] (Center) at (0,0) Emergent
Equivalence
Principle;
[->, thick] (Causality) – (Center); [->, thick] (RelationalTime) – (Center); [->, thick] (Trinity) – (Center);

Figure A.1: Three structural ingredients yielding the Equivalence Principle as an emergent low-energy symmetry.

A.7 The Equivalence Principle Reconsidered

A.7.1 EP variants and evidence

Distinguish weak equivalence (universality of free fall), Einstein equivalence (local Lorentz and position invariance), and strong equivalence (including gravitational self-energy). Precision tests impose tight bounds in many regimes [17].

A.7.2 EP as axiom in curvature-based gravity

In standard GR pedagogy, the EP motivates locally inertial frames that remove gravitational effects by coordinates, suggesting gravity is geometry rather than force. Powerful—but not unique—especially in metric–affine or gauge formulations [14].

A.7.3 EP as emergent in teleparallel and symmetric teleparallel

TEGR and STEGR separate inertial and gravitational contributions at the level of the connection, allowing EP-like phenomenology without taking the EP as fundamental. This supports an emergent reading tied to connection dynamics [14, 16, 15].

Remark A.7.1 (Didactic recap)**.** Quick map. *WEP: motion ∼ mass-independence; EEP: local nongravitational physics respects SR; SEP: even gravitational self-energy obeys equivalence. The present account preserves the tested content while shifting the interpretive load from axiom to emergence.*

A.8 Emergence of the EP from Relational Time and Connection Dynamics

A.8.1 Synthesis

Combine: (i) causality fixes the Lorentzian scaffold [8, 9]; (ii) relational constraints yield redshift and energy–energy interactions at low energy [20, 11]; and (iii) geometric codings are empirically equivalent [14, 16]. In this regime, the EP is an emergent organizing principle.

A.8.2 Low-energy universality

The effective coupling depends on energy operators rather than composition-specific constants. Universality of free fall and equality of inertial and gravitational mass then appear as consequences of the relational structure, not separate postulates [20, 14].

A.8.3 When deviations should appear

Deviations should track clock resolution, energy superpositions in the source, or departures from the perturbative regime. These deviations are coherence-dependent corrections to conditional evolution, not classical fifth forces, and they scale as ε_Λ^2 for systems approaching the perturbative boundary. Observable signatures include phase decoherence between non-identical quantum clocks and redshift modulations correlated with internal energy superpositions [18, 19].

Remark A.8.1 (Didactic recap)**.** One-sentence logic. *Causality* $\Rightarrow$ *Lorentz scaffold; relational time + small deformation* $\Rightarrow$ *redshift & energy–energy term; geometric trinity* $\Rightarrow$ *multiple codings; together* $\Rightarrow$ *EP as a low-energy symmetry with clear conditions for small deviations.*

The overall logical flow is depicted in Fig. A.2.

[every node/.style=draw, rectangle, rounded corners=4pt, align=center, font=, text width=3.8cm, minimum height=1.5cm, arrow/.style=->, thick]
(Causality) Operational Causality
(fixes Lorentzian kinematics); [below=1.2cm of Causality] (RelTime) Relational Time
+ Constraint Deformation; [right=0.8cm of Causality] (Trinity) Geometric Trinity:
Curvature $\leftrightarrow$ Torsion $\leftrightarrow$ Non-metricity; [below=1.2cm of Trinity] (EffHam) Emergent Redshift
+ Energy–Energy Coupling;
[right=0.8cm of Trinity, draw, thick, fill=gray!10] (EP) Emergent
Equivalence Principle;
[arrow] (Causality) – (RelTime); [arrow] (RelTime) – (EffHam);
[arrow] (EffHam) – (Trinity);
[arrow] (Trinity) – (EP);

Figure A.2: Logical flow from causal and relational structure to the emergence of the Equivalence Principle, shown as a two-row horizontal layout.

A.9 Phenomenology and Outlook

A.9.1 Consistency with weak-field tests

Standard redshift and Newtonian limits are recovered once Λ^{-1} is matched to the Newtonian potential scale [20, 17]. The geometric trinity ensures equivalent classical phenomenology across codings [14, 16].

A.9.2 Quantum-clock probes

Quantum-clock interferometry and related setups directly target coherence-dependent redshift corrections and offer clean tests of the relational mechanism [18, 19]. In particular, experiments comparing atomic or optical lattice clocks at different gravitational potentials could detect the predicted coherence-dependent phase lags once the relative clock energy scales approach
$epsilonL \sim 10^{-3}$. These effects would manifest as slight deviations from standard gravitational redshift that depend on clock-state superpositions rather than composition or mass. Modern quantum-clock interferometers already reach sensitivities where
$epsilonL \sim 10^{-3}$ is plausible. Comparing optical-lattice clocks at different potentials could reveal coherence-dependent phase lags as slight deviations from standard gravitational redshift—effects controlled by the ratio
$epsilonL = \|\hat{H}_S\|/\Lambda$.

A.9.3 Open problem: distance

A principled derivation of the $1/r$ form remains. Operational distance compatible with causal structure exists [8]; importing it into the relational constraint is a concrete next step.

Remark A.9.1 (Didactic recap)**.** For the experimental reader. *What's fixed now: leading redshift and Newtonian limits. What's to measure: coherence-dependent corrections. What's missing theoretically: an intrinsic route to* $1/r$ *from spatial reference frames within the same relational scaffold.*

A.10 Discussion

The account offered here keeps phenomenology intact while reordering foundations: causality sets the inertial scaffold; relational time and mild constraint deformations supply effective gravitational structure; and different geometric codings summarize the same empirical content. On this reading, the EP is a low-energy symmetry. The framework highlights coherence-dependent redshift as an actionable experimental target.

Looking ahead, quantum-clock interferometry and relativistic coherence tests provide the clearest path to probing the emergent regime described here. The present account thus positions the Equivalence Principle not merely as a cornerstone of geometry but as an experimentally accessible low-energy symmetry arising from relational dynamics amenable to falsification as clock coherence and energy resolution improve.

A.11 Conclusion

Operational causality fixes relativistic kinematics [8, 9]. Relational-time dynamics with weak deformations generate redshift and energy–energy interactions [20, 11]. The geometric trinity reveals multiple, inequivalent geometric summaries of identical phenomenology [14, 16]. Together these motivate treating the EP as emergent and identify quantum-clock tests as the right place to look for coherence-dependent corrections.

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Code availability

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Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

.1 Operator Domains and Regularization of Effective Evolution

The expression $\hat{H}_{\text{eff}} = \hat{H}_S(\hat{I} + \hat{H}_S/\Lambda)^{-1}$ requires care when $\hat{H}_S$ is unbounded or when the spectrum overlaps $-\Lambda$. In practice one can: (i) restrict to low-energy sectors $|\hat{H}_S| \ll \Lambda$ and use the expansion (A.3.3); (ii) model clocks/systems with bounded spectra; or (iii) introduce spectral cutoffs or smearing representing finite clock resolution. These standard choices make the relational evolution well-defined [20, 11].

.2 q–Desics in Quantum Gravity: Operator–Averaged Motion Beyond the Mean Metric

Orthodox Review with a Speculative Swirl–String Theory (SST) Appendix

In classical general relativity, freely falling test particles follow geodesics determined by the spacetime metric. In quantum-gravitational settings, it is common to approximate motion by geodesics of the expectation-value metric $\langle \hat{g}_{\mu\nu} \rangle$. Koch, Riahinia, and Rincón introduce *q–desics* as quantum-corrected analogs of geodesics, defined instead by the expectation value of an *affine-connection operator* $\langle \hat{\Gamma}^{\beta}_{\mu\nu} \rangle$. A central structural result is that q–desics coincide with geodesics of $\langle \hat{g} \rangle$ if and only if certain covariances between metric-derived operators vanish. This article reviews the framework in orthodox language, derives the covariance criterion, and summarizes implications for static, spherically symmetric quantum backgrounds. An appendix gives a clearly flagged, speculative mapping of the covariance structure to Swirl–String Theory (SST) as an example of how "beyond-mean-field" geometry can be encoded in second-moment microstructure data.

.3 Classical baseline: geodesics from an action

Let $(\mathcal{M}, g_{\mu\nu})$ be a Lorentzian spacetime and $x^\mu(\lambda)$ a worldline. For a massive particle, the reparameterization-invariant action can be written as

$$S[x; g] \;=\; -m \int d\lambda \, \sqrt{g_{\mu\nu}(x) \, \dot{x}^\mu \dot{x}^\nu}, \tag{.3.1}$$

where $\dot{x}^\mu \equiv dx^\mu/d\lambda$. Variation with respect to $x^\mu(\lambda)$ yields the geodesic equation

$$\ddot{x}^\beta + \Gamma^\beta_{\mu\nu}(g) \, \dot{x}^\mu \dot{x}^\nu = 0, \qquad \Gamma^\beta_{\mu\nu}(g) = \frac{1}{2} g^{\beta\alpha} (\partial_\mu g_{\nu\alpha} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu}). \tag{.3.2}$$

Null geodesics arise from the massless limit, together with the null constraint $g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0$.

.4 Common semiclassical proxy and its structural limitation

In many semiclassical or effective treatments, the quantum-gravitational background is summarized by an expectation value $\langle \hat{g}_{\mu\nu} \rangle$ and test-particle motion is approximated by geodesics of $\langle \hat{g} \rangle$. This is a *mean-field* replacement:

$$\Gamma^\beta_{\mu\nu}(g) \;\mapsto\; \Gamma^\beta_{\mu\nu}(\langle \hat{g} \rangle). \tag{.4.1}$$

A structural limitation is immediate: in general, *nonlinear* functions of operators do not commute with taking expectation values, so $\langle f(\hat{g}) \rangle \neq f(\langle \hat{g} \rangle)$ for generic states.

The q–desic framework makes this mismatch explicit at the level of the affine connection.

.5 q–Desics: definition via operator–level variational principle

Koch, Riahinia, and Rincón derive a quantum-corrected equation of motion by keeping the spacetime metric as an operator $\hat{g}_{\mu\nu}$ as long as possible, and taking the spacetime expectation value at the end. The resulting q–desic equation is (their Eq. (16))

$$\ddot{x}^\beta + \langle \hat{\Gamma}^\beta_{\mu\nu} \rangle \, \dot{x}^\mu \dot{x}^\nu = 0, \tag{.5.1}$$

with an operator-valued Christoffel structure whose expectation value is defined (their Eq. (17)) by

$$\langle \hat{\Gamma}^\beta_{\mu\nu} \rangle \;:=\; \left\langle \frac{1}{2} \left(\partial_\mu \hat{g}_{\nu\alpha} + \partial_\nu \hat{g}_{\mu\alpha} - \partial_\alpha \hat{g}_{\mu\nu} \right) \hat{g}^{\alpha\beta} \right\rangle. \tag{.5.2}$$

Here $\hat{g}^{\alpha\beta}$ denotes an inverse-metric operator, assumed to be well-defined in the chosen quantum-gravity setting. For massive vs. massless motion, the usual normalization constraint is replaced by the expectation-value constraint (their Eq. (25))

$$\langle \hat{g}_{\mu\nu} \rangle \, u^\mu u^\nu = \begin{cases} 1, & \text{massive}, \\ 0, & \text{massless}, \end{cases} \qquad u^\mu \equiv \dot{x}^\mu. \tag{.5.3}$$

Interpretation. Equation (.5.1) shows that the "effective geometry felt by a test particle" can depend on $\langle \hat{\Gamma} \rangle$ rather than solely on $\Gamma(\langle \hat{g} \rangle)$.

.6 Covariance criterion: when q–desics reduce to g–eodesics

Define the *geodesic-of-the-mean* connection

$$\Gamma^{\beta}_{\mu\nu}(\langle\hat{g}\rangle) = \frac{1}{2}\,\langle\hat{g}\rangle^{\beta\alpha}\big(\partial_\mu\langle\hat{g}_{\nu\alpha}\rangle + \partial_\nu\langle\hat{g}_{\mu\alpha}\rangle - \partial_\alpha\langle\hat{g}_{\mu\nu}\rangle\big), \tag{.6.1}$$

and the *q–desic covariance* (their Eq. (81)) as the difference

$$\mathrm{Cov}^{\beta}{}_{\mu\nu}(\Gamma) \;:=\; \langle\hat{\Gamma}^{\beta}_{\mu\nu}\rangle - \Gamma^{\beta}_{\mu\nu}(\langle\hat{g}\rangle). \tag{.6.2}$$

Introduce the linear-in-derivatives operator (their Eq. (82))

$$\hat{X}_{\mu\nu\alpha} := \frac{1}{4}\,(\partial_\mu\hat{g}_{\nu\alpha} + \partial_\nu\hat{g}_{\mu\alpha} - \partial_\alpha\hat{g}_{\mu\nu}). \tag{.6.3}$$

Then the covariance can be rewritten schematically as

$$\mathrm{Cov}^{\beta}{}_{\mu\nu}(\Gamma) = \Big\langle 2\,\hat{X}_{\mu\nu\alpha}\,\hat{g}^{\alpha\beta}\Big\rangle - 2\,\Big\langle\hat{X}_{\mu\nu\alpha}\Big\rangle\,\Big\langle\hat{g}^{\alpha\beta}\Big\rangle \;+\; \text{(ordering/commutator terms)}, \tag{.6.4}$$

i.e. it measures the failure of factorization of the mixed operator product. With Weyl ordering in observables, the remaining contribution can be expressed in terms of an anticommutator expectation (their Eq. (83)):

$$\mathrm{Cov}^{\beta}{}_{\mu\nu}(\Gamma) = \Big\langle\!\Big\langle \{\hat{X}_{\mu\nu\alpha},\, \hat{g}^{\alpha\beta}\}\Big\rangle\!\Big\rangle\,, \tag{.6.5}$$

where the double-bracket notation indicates "connected" (covariance-like) parts.

Consequence. If $\mathrm{Cov}^{\beta}{}_{\mu\nu}(\Gamma) = 0$ for a given state, then

$$\langle\hat{\Gamma}^{\beta}_{\mu\nu}\rangle = \Gamma^{\beta}_{\mu\nu}(\langle\hat{g}\rangle), \tag{.6.6}$$

and q–desics coincide with geodesics of the mean metric. Otherwise, they differ.

.7 Static spherically symmetric quantum backgrounds (orthodox summary)

The paper applies (.5.1) to static, spherically symmetric quantum-gravitational backgrounds obtained via canonical quantization under a static equal-radius (SER) condition. In standard coordinates, one may parameterize a classical static, spherically symmetric line element as

$$ds^2 = -n(r)^2\,dt^2 + g(r)^2\,dr^2 + r^2(d\theta^2 + \sin^2\theta\,d\phi^2), \tag{.7.1}$$

and promote the radial functions to operators $\hat{n}(r)$, $\hat{g}(r)$. Inputs to q–desic motion are then expectation values such as $\langle\hat{g}\rangle$, $\langle\hat{n}^2\rangle$, and mixed products entering (.5.2).

A representative example of the structure of these expectation values (quoted in the paper's "selected expectation values" discussion) is an expansion of the form

$$\langle\hat{g}\rangle = (1 + \epsilon_{0,0}) + \frac{C_{1,0}}{r} - \frac{\Lambda(1 + \epsilon_{0,0})}{3}\,r^2 + \cdots\,, \tag{.7.2}$$

with state-dependent integration constants (e.g. $\epsilon_{i,j}$) and couplings (e.g. $C_{1,0}$, Λ). The q–desic observables depend not only on (.7.2) but also on covariances such as (.6.2)–(.6.5), which encode additional state information beyond the mean fields.

.8 Observable deviations: what changes in principle?

Two broad classes of effects follow structurally:

.8.1 Radial null motion and "horizons"

For radial null propagation, a classical "horizon" location is typically inferred from the vanishing of an effective lapse function or from divergences in dr/dt depending on coordinates. In q–desic motion, the relevant relation involves $\langle\hat{\Gamma}\rangle$ and can shift the location where coordinate speeds change qualitatively, even if $\langle\hat{g}\rangle$ resembles a classical form. Operationally, horizon-identification becomes *state-dependent* through covariances.

.8.2 Circular motion of massive particles

For circular motion, GR yields the standard relation between orbital velocity, enclosed mass, and radius in the weak-field regime. In the q–desic framework, the orbital relations receive corrections controlled by: (i) deviations in mean metric expectation values; and (ii) non-factorizing mixed correlators (the covariance terms). Thus, even far above the Planck scale, small but structured deviations can arise if the background quantum state has nontrivial correlations.

.9 Relation to other orthodox frameworks (brief)

The paper emphasizes analogies to standard operator-vs-average issues (e.g. the Ehrenfest theorem in quantum mechanics), and notes links to:

- **Stochastic gravity:** where induced and intrinsic metric fluctuations are characterized by noise kernels and correlators;
- **Effective average action / FRG:** where scale-dependent effective metrics $\langle\hat{g}\rangle_k$ arise from $\Gamma_k[g]$;
- **Non-geodesic motion in GR:** e.g. Mathisson–Papapetrou–Dixon motion of spinning bodies (classical non-geodesicity from spin-curvature).

.10 Conclusions (orthodox)

q–desics provide a clean operator-defined extension of classical geodesic motion, replacing $\Gamma(\langle\hat{g}\rangle)$ by $\langle\hat{\Gamma}\rangle$. The key structural object is the covariance (.6.2), which measures the failure of factorization of mixed metric-derived operators. Consequently, two quantum states with identical $\langle\hat{g}\rangle$ may still yield different test-particle motion if their covariances differ. This isolates precisely what additional geometric information beyond the mean metric is required to predict motion in quantum spacetimes.

.1 Speculative SST appendix: mapping the covariance structure to swirl microstructure

Scope and warning

This appendix is **not** orthodox GR/QG. It is a **speculative** structural translation into Swirl–String Theory (SST) language, intended to extract a reusable lemma:

> *Mean-field "geometry" is insufficient when motion depends on mixed correlators; one must model second moments of the microstructure.*

.1.1 Minimal dictionary (structural, not ontological)

We treat the q–desic covariance as a template for how a coarse-grained continuum description can miss state information. A minimal SST-leaning dictionary is:

q–desic/QG object	SST structural analog (hypothesis)
$\langle \hat{g}_{\mu\nu} \rangle$	coarse-grained foliation/clock-effective metric
$\langle \hat{\Gamma}^{\beta}_{\mu\nu} \rangle$	coarse-grained affine response of the medium
$\mathrm{Cov}(\Gamma)$	second-moment microstructure data (mixed correlators)

.1.2 A concrete toy translation: "inverse-times-gradient" mismatch

In SST one often introduces a *clock field* (time-scaling functional) built from a local medium variable (e.g. a swirl-clock proxy). Abstractly, suppose an effective lapse-like factor $N(x)$ is a nonlinear functional of a micro-variable $\chi(x)$, and appears in an effective metric component $g_{tt} = -N^2$. Then the classical connection component relevant for radial acceleration contains

$$\Gamma^r{}_{tt} \sim -\frac{1}{2}\, g^{rr}\, \partial_r(N^2). \tag{.1.1}$$

Promoting N and g^{rr} to micro-operators (or random fields) and coarse-graining suggests two inequivalent closures:

$$\text{(mean-metric closure)} \quad \Gamma^r{}_{tt}(\langle g \rangle) \;\propto\; \langle g^{rr} \rangle\, \partial_r \langle N^2 \rangle, \tag{.1.2}$$

$$\text{(q–desic-like closure)} \quad \langle \Gamma^r{}_{tt} \rangle \;\propto\; \Big\langle g^{rr}\, \partial_r(N^2) \Big\rangle \;=\; \langle g^{rr} \rangle\, \partial_r \langle N^2 \rangle \;+\; \mathrm{Cov}(g^{rr}, \partial_r(N^2)). \tag{.1.3}$$

Equation (.1.3) is the SST-side moral equivalent of (.6.2): *the correction is controlled by a mixed correlator, not by the mean fields alone.*

.1.3 Numerical scale note using SST constants

If a toy clock functional depends on a characteristic swirl speed scale $\mathbf{v}_{\circlearrowleft}$ via a dimensionless ratio $(\mathbf{v}_{\circlearrowleft}/c)^2$, then the small parameter is

$$\left(\frac{\mathbf{v}_{\circlearrowleft}}{c}\right)^2 = \left(\frac{1.09384563 \times 10^6 \text{ m/s}}{2.99792458 \times 10^8 \text{ m/s}}\right)^2 \approx 1.3313 \times 10^{-5}. \tag{.1.4}$$

This does *not* by itself fix the covariance magnitude, but it provides an SST-consistent reference scale for any clock-field nonlinearity expanded in $(\mathbf{v}_{\circlearrowleft}/c)^2$.

.1.4 Falsifiable SST-facing lesson

If SST predicts that macroscopic motion is captured by a mean-field foliation alone, then it corresponds to the constraint

$$\mathrm{Cov}(g^{rr}, \partial_r(N^2)) \approx 0 \tag{.1.5}$$

for relevant states of the medium. Conversely, if SST allows persistent correlated microstructure, then deviations from mean-field motion become possible *even when mean fields are unchanged.* This is a precise place to attach experimental or observational constraints: they target *correlations*, not merely averages.

10-year-old analogy

If you only look at the *average* speed of a river you might think a boat goes straight, but if fast swirls tend to happen exactly where the water is also shallow, the boat gets pushed in a special way. The pushing depends on how those two things *line up together*, not just on their averages.

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Appendix A

SST-66 — Relational time and intrinsic temporal stochasticity

Time from Swirl: A Hydrodynamic Origin of Proper Time

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April 19, 2026

Abstract

Swirl-String Theory reformulates physical time as a relational observable associated with a conserved event current rather than as an external parameter. We explore the phenomenological consequence that clock readouts may exhibit intrinsic temporal broadening, independent of environmental noise or relativistic time dilation. We introduce an operational ansatz for time-of-arrival fluctuations and discuss how such effects could scale with gradients of the clock foliation field. Finally, we outline exploratory clock-comparison experiments capable of constraining this temporal stochasticity and discuss implications for quantum measurement and decoherence.

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A.1 Relational Time in Swirl–String Theory

In Swirl–String Theory (SST), physical time is not introduced as a fundamental external parameter. Instead, time is defined operationally through a conserved event current associated with the underlying foliation field. Clock readings correspond to the accumulation of discrete events along integral curves of a preferred timelike vector field u^μ.

This construction is fixed canonically in SST–31 and Canon v0.7.7 and is equivalent in the infrared to hypersurface–orthogonal Einstein–Æther (khronometric) theory. No stochasticity is postulated at the level of the action.

A.2 Intrinsic Temporal Broadening

While the event current is conserved, its operational realization in physical clocks involves counting discrete microscopic processes. As a consequence, clock readouts need not be perfectly sharp even in the absence of environmental noise.

We define *intrinsic temporal broadening* as a fundamental variance in time–of–arrival measurements that persists after all known classical, thermal, quantum, and relativistic corrections are removed.

Operationally, the observed time–of–arrival distribution $P_{\rm obs}(\Theta)$ is modeled as

$$P_{\rm obs}(\Theta) = \int dt\, P_{\rm cl}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right], \tag{A.2.1}$$

where σ_τ^2 parametrizes intrinsic temporal stochasticity.

A.3 Scaling Ansatz and Physical Interpretation

Canon v0.7.7 implies that relational time is defined relative to the clock–foliation field. Accordingly, intrinsic temporal broadening may depend on spatial inhomogeneities of the foliation.

We therefore adopt the phenomenological ansatz

$$\sigma_\tau^2 = \sigma_\tau^2(\nabla\chi), \tag{A.3.1}$$

where χ denotes the scalar foliation (clock) potential.

Physically, this dependence reflects fluctuations in the local event rate induced by gradients of the clock field. No claim is made regarding the microscopic origin or spectral structure of this noise, which remains an open parameterization.

A.4 Experimental Search Channels and Falsification

Intrinsic temporal broadening predicts observable consequences only in high–precision differential clock experiments. Suitable search channels include:

- co–located optical lattice clocks operated in differential mode,
- entangled clock pairs sensitive to phase decoherence,
- controlled modulation of local clock–field gradients.

A null result at sensitivity $\sigma_\tau \lesssim 10^{-19}\,\mathrm{s}/\sqrt{\mathrm{Hz}}$ places direct bounds on the admissible stochastic sector of SST. Conversely, observation of clock decoherence correlated with $\nabla\chi$ would constitute evidence for relational time fluctuations distinct from standard quantum or gravitational noise.

This work therefore defines a falsifiable experimental program rather than a prediction of detectability.

A.5 Discussion and Outlook

A.5.1 Summary of Results

This work reformulates physical time in Swirl–String Theory (SST) as a relational observable associated with a conserved event current rather than as an external evolution parameter. Within this framework, clock readouts are not assumed *a priori* to correspond to a perfectly smooth variable, even in isolated systems.

We introduced an operational ansatz for intrinsic temporal broadening, modeled as a convolution of ideal clock evolution with a stochastic kernel characterized by a variance σ_τ^2. This quantity parametrizes intrinsic clock noise that is conceptually distinct from environmental disturbances, relativistic time dilation, or measurement backaction.

No claim has been made regarding the magnitude or detectability of σ_τ^2 with existing technology. The purpose of this work is to define a consistent phenomenological framework and an associated experimental search channel.

A.5.2 Relation to Other SST Results

The temporal stochasticity discussed here is logically independent of velocity-dependent mass effects and preferred-frame corrections addressed elsewhere. In particular, gravity-sector tests of foliation-referenced mass anisotropy are treated separately and do not enter the present analysis.

Accordingly, this work should be read as complementary to, but distinct from, operational follow-ups of the SST mass functional. The results presented here depend only on the relational definition of time fixed in the SST Canon and do not modify the gravitational sector.

A.5.3 Falsification and Outlook

The framework presented admits clear falsification pathways.

A null result in high-precision differential clock experiments constrains the allowed stiffness and smoothness of the SST clock foliation. Conversely, observation of excess decoherence or time-of-arrival broadening that scales with controlled gradients of the clock field, while surviving standard noise rejection protocols, would provide evidence for intrinsic temporal stochasticity.

Future work may refine the spectral properties of σ_τ^2, investigate its coupling to quantum coherence, and explore its implications for foundational questions in quantum measurement and time observables.

The present analysis establishes the minimal phenomenological groundwork required for such investigations, without extending beyond empirically testable statements.